

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)			A [changing] current in the base or primary coil (1) sets up a [changing] magnetic field in the <u>iron core</u> (1) If there is no reference to either a changing current or changing magnetic field award 1 of the first 2 marks only <u>induces</u> a voltage (accept current) in the secondary coil (1) Don't accept cutting magnetic flux lines	3			3		
	(b)	(i)		The [electrical] energy (or power) [don't accept electricity] (1) is not lost or wasted [as heat] / supplied to the primary coil is totally passed through to the secondary coil (1) Or total energy in = [electrical or] useful energy out (2) Or power in = power out (2)	2			2		
		(ii)		$\frac{V_1}{V_2} = \frac{N_1}{N_2}$ so $\frac{230}{V_2} = \frac{9200}{480}$ (1) for substitution Therefore $V_2 = \frac{230 \times 480}{9200} = 12$ [V] (1)	1	1		2	2	
		(iii)		Selection of : $P = V \times I$ or by implication (1) Substitution and conversion: 12 (ecf) $\times 0.1$ (1) $= 1.2$ [W] (1) Answer of 1.2×10^n award 2 marks	1	1 1		3	2	
				Question 3 total	7	3	0	10	4	0